FP 2:UFO Sightings and Machine Learning for National Security

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Introduction

Reports of Unidentified Flying Object (UFO) sightings have usually been dismissed as cases of misidentifications of natural phenomena, aircrafts or hoaxes. While reports of UFOs may not capture the attention of scientists, government agencies and officals might start looking to take them seriously.

To effecienly process, filter, and identify the thousands of UFO sighting reports, our goal is to examine which text analysis approaches and machine learning techniques can be used to aid government agencies and officals in their elimination process of non-threats, such as weather balloons, natural phenomena, or commerical aircrafts. We also want to change the historical view of UFO Sightings reports as “dramatic accounts” to indications of security concerns.

Data

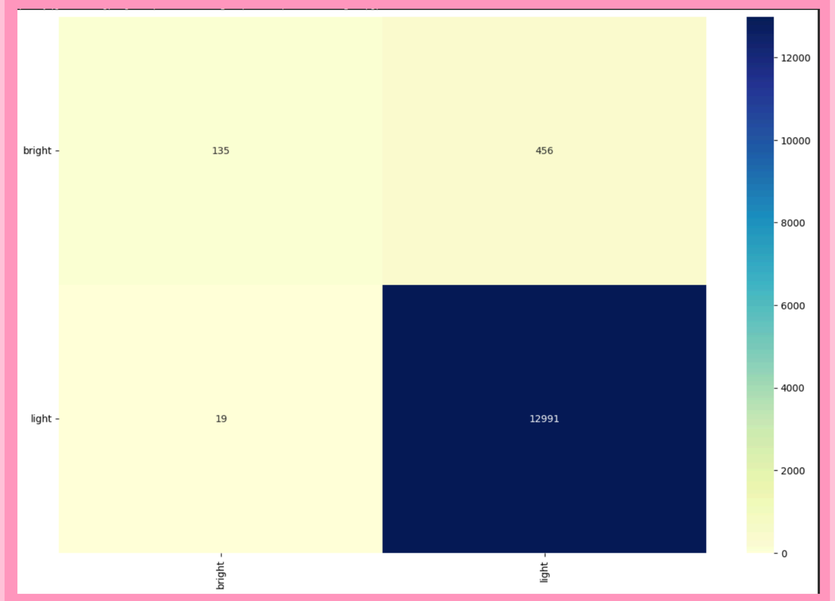
- Sourced from the Curated Dataset from the TidyTuesday project GitHub repo by textmining-infopros
- Total reports: 80,332 UFO sightings

Methodology

- Missing Values “NaN’ were replaced with missing strings.
- Sentiment analysis to examine the emotions experienced by the witnesses.
- Topic modeling with Bag-of-Words and LDA Models to determine what the “UFOs” could really be.
- Support Vector Machine model to examine descriptions of “light” and “bright” UFOs in the United States and Canada.

	sentiment_score
count	102678.000000
mean	0.087223
std	0.280808
min	-0.913600
25%	0.000000
50%	0.000000
75%	0.340000
max	0.941100

Figure 1: Sentiment Scores



Support Vector Machine classified the features and targets with a 94.4% accuracy.

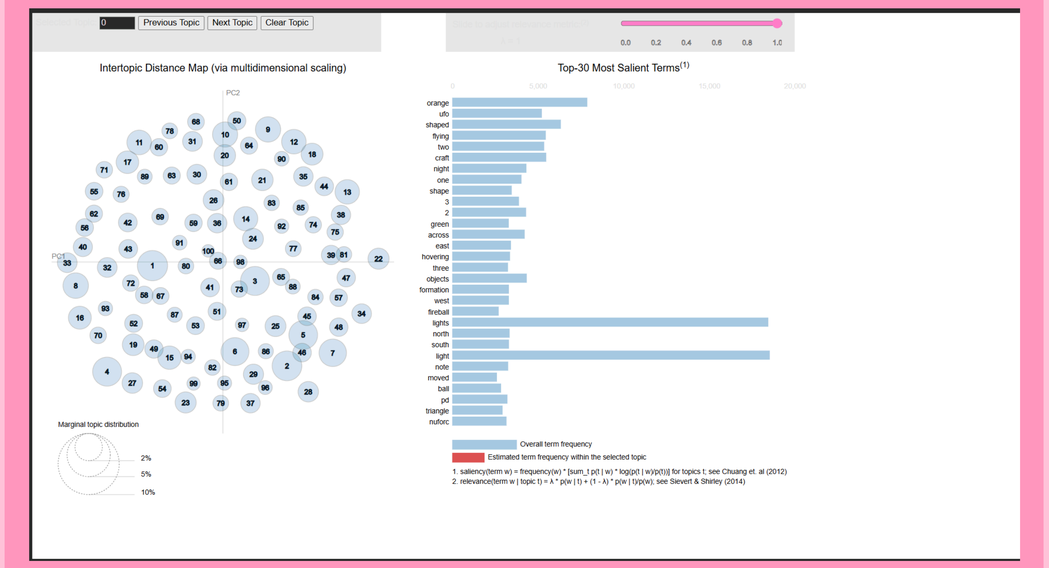


Figure 2: All 100 Topics Plotted on a Two-Dimensional Space

Findings

- The mean value of +0.08 indicates an overall neutral to slightly positive sentiment, which is most likely reflecting feelings of awe or confusion over what reporters witnessed.
- Topic Modeling revealed high recurring words of “light/lights”, “orange”, and “shaped”
- A 94.4% accuracy rate from the Support Vector Machine classifier reveals 12991 correctly classified US descriptions, and 135 correctly CA descriptions. Useful for geo mapping.

Conclusion

Using the “UFO Sightings around the world” dataset for textual analysis and machine learning provided us with a starting point towards uncovering a potential national security threat.

With our results from the Support Vector Machine model, we can use this information for further analysis over the exact geo location of where the bright lights occured. Perhaps there are important military bases nearby.

References

https://github.com/rfordatascience/tidytuesday/tree/main/data/2019/2019-06-25

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